

GenAI Final Project Comprehensive Guide

5 Project Options • Detailed Breakdown • 3-Person Work Distribution

Advanced Topics in Generative AI (MAIB AI 220)
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Introduction & How to Use This Guide

This guide presents five carefully designed final project options for the Advanced Topics in Generative AI course (MAIB AI 220). Each project has been selected to align with the course learning outcomes and the grading rubric, while being equally and fairly divisible between a team of three members.

For every project you will find: a full conceptual explanation, the course topics it covers, a detailed scope and methodology, a clear 3-person work breakdown, expected deliverables, and a realistic time estimate.

Colour Key for Work Distribution Tables

Person A	Person B	Person C
Data & Infrastructure	Core AI / Modelling	Evaluation, Presentation & Demo

Across all projects Person A focuses on data pipelines and infrastructure, Person B owns the core modelling and AI implementation, and Person C handles evaluation, explainability, and the final presentation. This split ensures each member has a distinct and equally weighted contribution.

■ PROJECT 1 — TOP RECOMMENDATION

Agentic RAG System for a Real Business Domain

Build an LLM-powered agent that retrieves, reasons, and explains answers over a domain-specific corpus.

COURSE WEEKS	DIFFICULTY	DURATION
Weeks 2, 5 & 6 — RAG, LLM Agents, LLM Explainability	High	5–6 weeks

Project Overview

Retrieval-Augmented Generation (RAG) is one of the most impactful techniques in enterprise AI today. Rather than relying purely on a model's parametric knowledge, RAG grounds responses in real documents — making the system accurate, up-to-date, and trustworthy. An Agentic RAG system goes one step further: it wraps the retrieval-generation loop inside an autonomous agent that can decide which documents to retrieve, how many retrieval hops to perform, and when to stop. This project implements such a system on a real-world domain corpus (e.g., company annual reports, healthcare guidelines, or legal contracts) and adds an explainability layer so users understand why the agent produced a given answer.

Key Concepts Covered

- Transformer-based dense retrieval: encoding queries and documents into a shared embedding space so semantic similarity can be computed via cosine or dot-product distance.
- Vector databases (FAISS, Chroma, Pinecone): indexing high-dimensional embeddings for sub-millisecond approximate nearest-neighbour search at scale.
- RAG pipeline design: chunking strategy (fixed-size vs. semantic), overlap, metadata filtering, re-ranking with a cross-encoder, and context injection into the prompt.
- LLM Agent frameworks (LangChain, LlamaIndex, AutoGen): tool calling, chain-of-thought orchestration, multi-step reasoning, and self-reflection loops.
- Explainability: attribution of the model's output back to retrieved document spans, uncertainty quantification, and confidence scores for end-user transparency.

Step-by-Step Methodology

- Select a domain corpus (minimum 500 documents). Parse, clean, and chunk documents.
- Embed chunks using a pre-trained encoder (e.g., text-embedding-3-small or bge-large).
- Store embeddings in a vector store; implement metadata-aware retrieval.
- Build an agent loop: ReAct-style reasoning, tool calls for retrieval, answer synthesis.
- Add an explainability layer: highlight source passages, display confidence, log reasoning steps.

6. Evaluate with RAGAS metrics: faithfulness, answer relevancy, context precision & recall.

Expected Deliverables

- ✓ Working end-to-end demo (Gradio or Streamlit UI).
- ✓ Evaluation report with RAGAS scores benchmarked against a naive RAG baseline.
- ✓ Explainability dashboard showing source attribution per answer.
- ✓ 10-minute classroom presentation with live demo.

3-Person Work Distribution

Person A	Person B	Person C
Data pipeline: corpus collection, cleaning, chunking, embedding, vector store indexing, and retrieval evaluation.	Agent framework: ReAct loop, tool definitions, multi-hop reasoning, self-reflection, LLM integration.	Explainability + presentation: source attribution, RAGAS evaluation, Streamlit UI, slides, and live demo.

Why this impresses the professor: This project directly maps to three course weeks simultaneously and produces original evaluation results. The combination of agentic reasoning and explainability is precisely what the professor highlights in the rubric under 'Effective application of deep learning techniques' and 'In-depth analysis'. Enterprise RAG is also a hot topic in industry, making the business relevance immediately obvious.

■ PROJECT 2

Multimodal Product Intelligence System

A vision-language AI that reads product images and text to generate insights and recommendations.

COURSE WEEKS	DIFFICULTY	DURATION
Weeks 7, 9 & 11 — Vision Models, Multimodal LLMs, Recommendation Systems	High	5 weeks

Project Overview

Multimodal AI — systems that process both images and text together — is one of the fastest-growing areas in generative AI research. This project builds a product intelligence system that ingests a catalogue of product images alongside textual descriptions and uses a multimodal foundation model (e.g., LLaVA, GPT-4o Vision, or BLIP-2) to perform cross-modal reasoning. The system will generate product summaries, answer natural language queries about products ('Which laptop has the best port selection?'), and produce personalised recommendations based on user preferences expressed in free text.

Key Concepts Covered

- Vision Transformers (ViT): patch-based tokenisation of images, positional embeddings, and self-attention over visual tokens — mirroring the NLP Transformer architecture.
- Contrastive pre-training (CLIP): aligning image and text embeddings in a shared latent space via contrastive loss, enabling zero-shot image-text matching.
- Multimodal fusion: early fusion (concatenating modalities at input), late fusion (combining modality-specific representations), and cross-attention fusion used in models like Flamingo and LLaVA.
- Visual instruction tuning: fine-tuning a language model to follow instructions grounded in image observations, enabling dialogue about visual content.
- Foundation model-based recommendation: reprogramming an LLM to serve as a recommendation engine by encoding user preferences and item features as natural language.

Step-by-Step Methodology

1. Assemble a product dataset with images + text metadata (e.g., Amazon product catalogue subset).
2. Embed images with CLIP or ViT; embed text with a sentence encoder.
3. Build a cross-modal retrieval pipeline: query by text, retrieve visually and semantically similar products.
4. Integrate a multimodal LLM for generative responses and product comparisons.
5. Implement a recommendation module using preference description as the query.

6. Evaluate with retrieval metrics (Recall@K) and user-study scoring on recommendation quality.

Expected Deliverables

- ✓ Image + text dual-encoder retrieval system with benchmarked Recall@5 and Recall@10.
- ✓ Multimodal QA interface: ask questions about any product in natural language.
- ✓ Personalised recommendation engine demo.
- ✓ Presentation with visual results and ablation study (text-only vs. multimodal).

3-Person Work Distribution

Person A	Person B	Person C
Dataset assembly, image preprocessing, CLIP/ViT embedding pipeline, and vector store for cross-modal retrieval.	Multimodal LLM integration (LLaVA or BLIP-2), visual instruction tuning or prompting, QA interface.	Recommendation module, retrieval evaluation metrics, ablation study, UI/demo, and final presentation.

Why this impresses the professor: Multimodal AI is explicitly covered in Week 9 of the curriculum and is one of the hottest research areas. Presenting a working demo that visually shows cross-modal understanding will be immediately captivating in a classroom setting. The ablation study (comparing text-only vs. multimodal) adds the research rigour that the professor's rubric rewards under 'Depth' and 'Content Quality'.

PROJECT 3

LLM Benchmark & Comparative Evaluation Study

Design a novel evaluation axis and rigorously benchmark 4 state-of-the-art LLMs on tasks the field hasn't tested.

COURSE WEEKS	DIFFICULTY	DURATION
Weeks 1, 3 & 4 — Language Model Advances, Scaling, Chain-of-Thought	Medium-High	4–5 weeks

Project Overview

Rather than building a single system, this project contributes original empirical knowledge by conducting a rigorous, publication-quality benchmark study. The team selects a task dimension that existing benchmarks underserve — such as multi-hop scientific reasoning, long-form structured output quality, multilingual business communication, or chain-of-thought faithfulness — and systematically evaluates four frontier models (e.g., GPT-4o, Gemini 1.5 Pro, Claude 3 Opus, Llama 3 70B) on a carefully curated dataset. The project produces original findings that can be directly referenced in the research paper assessment.

Key Concepts Covered

- LLM evaluation methodology: designing prompts that isolate a single capability, controlling for confounders like prompt sensitivity and ordering bias.
- Benchmark dataset construction: sourcing and curating examples, annotating ground truth, ensuring coverage and difficulty calibration.
- Chain-of-Thought evaluation: assessing not just final answer accuracy but intermediate reasoning faithfulness — do the reasoning steps actually support the conclusion?
- Scaling laws: empirically verifying or challenging published claims about how performance scales with model size on the chosen task dimension.
- Statistical analysis: inter-annotator agreement, bootstrap confidence intervals, significance testing, and failure mode categorisation.

Step-by-Step Methodology

1. Define the evaluation dimension and justify its importance with a literature review.
2. Curate or construct a dataset of 200–500 examples with gold-standard annotations.
3. Design a prompt suite: zero-shot, few-shot (3-shot, 5-shot), and chain-of-thought variants.
4. Run all prompts on 4 selected LLMs via their APIs; log all outputs systematically.
5. Score outputs automatically where possible; use human annotation for open-ended tasks.
6. Perform statistical analysis; visualise results with radar charts, bar plots, and error analyses.

Expected Deliverables

- ✓ Benchmark dataset (released publicly on Hugging Face if possible — bonus marks!)
- ✓ Full evaluation report: methodology, results tables, statistical significance, and failure analysis.
- ✓ Visualisation dashboard showing model-by-model performance breakdown.
- ✓ Presentation framed as a research paper walkthrough.

3-Person Work Distribution

Person A	Person B	Person C
Literature review, dataset curation/construction, gold-standard annotation, and prompt design.	API integration for all 4 LLMs, systematic evaluation runs, output logging, and automated scoring.	Human evaluation protocol, statistical analysis, visualisations, report writing, and presentation.

Why this impresses the professor: Professors are impressed most by original findings — and this project generates them. The evaluation rubric explicitly rewards 'highly innovative topic choice' and 'extensive research with comprehensive insights'. A benchmark study that produces publishable results is the strongest possible demonstration of research methods competence (CLO5), the hardest course learning outcome to evidence.

PROJECT 4

Text-to-Video Prompt Engineering Study

Systematically analyse how prompt design controls video generation quality and build a reusable prompt taxonomy.

COURSE WEEKS	DIFFICULTY	DURATION
Week 8 — Video Generation Models, Diffusion Transformers	Medium	4 weeks

Project Overview

Video generation is the newest frontier of generative AI, with models like Sora, Kling, Runway Gen-3, and CogVideoX emerging in 2024. Unlike image generation, video requires temporal coherence — the model must maintain object identity, motion consistency, and scene continuity across hundreds of frames. This project conducts a structured empirical study of how prompt engineering strategies (motion descriptors, camera directives, style tokens, temporal cues, and negative prompts) affect the semantic accuracy, motion quality, and aesthetic coherence of generated videos. The output is a reusable prompt taxonomy and best-practice guide — a practical contribution with real value for practitioners.

Key Concepts Covered

- Diffusion models for video: extending denoising diffusion probabilistic models (DDPM) to the temporal dimension — treating a video as a sequence of latent frames and denoising across both space and time.
- Diffusion Transformers (DiT): replacing the U-Net backbone of diffusion models with a Vision Transformer, enabling better scalability and quality at high resolutions — the architecture underlying Sora.
- Temporal attention mechanisms: self-attention applied across the time axis so each frame can attend to all other frames, enforcing temporal consistency.
- Prompt engineering theory: semantic precision, compositional prompting, style token injection, and the role of negative prompts in classifier-free guidance.
- Video quality metrics: FVD (Frechet Video Distance), CLIPSIM for text-video alignment, human preference scoring (ImageReward analogues for video).

Step-by-Step Methodology

1. Literature review of text-to-video architectures and existing prompt studies.
2. Design a systematic prompt experiment: vary one prompt dimension at a time (motion, camera, style, length, negative) while holding others constant.
3. Generate 150–200 videos across 4–5 models (Runway, Kling, CogVideoX, etc.).
4. Score all videos on: text alignment, motion smoothness, temporal consistency, and aesthetic quality.

5. Build a prompt taxonomy: categorise prompt patterns by the quality dimension they most influence.
6. Produce a best-practice guide with before/after examples.

Expected Deliverables

- ✓ Prompt taxonomy document classifying 40+ prompt patterns by effect.
- ✓ Quantitative scoring tables and quality heatmaps across models and prompt types.
- ✓ A curated demo reel of best/worst generated videos illustrating key findings.
- ✓ Presentation with live generation demo.

3-Person Work Distribution

Person A	Person B	Person C
Literature review on video diffusion architectures (DiT, temporal attention) and existing prompt studies. Define the experimental design.	Run all video generation experiments across all models; maintain systematic logging of prompts, parameters, and outputs.	Scoring protocol, quantitative analysis, taxonomy construction, demo reel editing, and final presentation.

Why this impresses the professor: Week 8 (Video Generation) is niche enough that very few student groups will go deep here, giving the team an immediate differentiation advantage. A live in-class demo showing generated videos is visually stunning and memorable. The systematic experimental design also demonstrates research rigour that aligns directly with the professor's rubric criteria.

PROJECT 5

Domain-Specific Foundation Model Fine-Tuning with PEFT

Fine-tune a foundation model on a niche scientific or business dataset using parameter-efficient methods and benchmark it.

COURSE WEEKS	DIFFICULTY	DURATION
Weeks 2, 10 & 11 — Fine-Tuning, Foundation Models for Science & Emerging Domains	Medium-High	5 weeks

Project Overview

Large foundation models are powerful generalists, but domain-specific tasks — predicting protein function, forecasting financial time series, or classifying medical notes — often require adapting them to specialised vocabulary, data distributions, and task formats. Parameter-Efficient Fine-Tuning (PEFT) methods like LoRA (Low-Rank Adaptation) and QLoRA make this feasible without requiring the enormous compute of full fine-tuning: they freeze the base model and insert small trainable adapter layers, reducing trainable parameters by up to 99%. This project applies PEFT to fine-tune a mid-sized foundation model (7B–13B parameters) on a domain dataset and rigorously evaluates the fine-tuned model against the base model and a larger general model.

Key Concepts Covered

- Parameter-Efficient Fine-Tuning (PEFT): methods that adapt large pre-trained models with minimal compute by training only a small fraction of parameters — including LoRA, QLoRA, prefix tuning, and prompt tuning.
- LoRA (Low-Rank Adaptation): injecting trainable low-rank decomposition matrices (A and B, rank r) into the attention weight matrices, so the weight update $\Delta W = BA$ can be learned cheaply and merged at inference.
- QLoRA: combining 4-bit NF4 quantisation of the base model with LoRA adapters, reducing VRAM requirements to fine-tune a 13B model on a single consumer GPU.
- Domain adaptation challenges: vocabulary shift (out-of-vocabulary domain terms), distributional shift (domain text statistics differ from pre-training data), and task format alignment.
- Foundation models for science / time series: how domain-specific pre-training objectives (e.g., masked atom prediction in chemistry, patch-based time series masking) differ from NLP pre-training.

Step-by-Step Methodology

1. Select a domain dataset (e.g., PubMedQA for biomedical, FinBench for finance, ECG/EHR for healthcare).

2. Establish a baseline: evaluate the off-the-shelf base model (e.g., Llama 3 8B) on the target task.
3. Fine-tune with QLoRA using Hugging Face PEFT + TRL libraries; document hyperparameter choices.
4. Run ablations: compare LoRA rank $r=4, 8, 16$; compare PEFT vs. full fine-tune on a smaller model.
5. Evaluate fine-tuned model on held-out test set; compare to base model and a larger general model (GPT-4o).
6. Analyse failure modes: where does fine-tuning help most and least?

Expected Deliverables

- ✓ Fine-tuned model weights (shared on Hugging Face Hub — shows real-world impact).
- ✓ Ablation study: performance vs. trainable parameter count curve.
- ✓ Evaluation report comparing base, fine-tuned, and general frontier model.
- ✓ Presentation with clear before/after output examples and training loss curves.

3-Person Work Distribution

Person A	Person B	Person C
Dataset sourcing, preprocessing, tokenisation, train/val/test splits, and baseline evaluation of the off-the-shelf model.	QLoRA/LoRA fine-tuning pipeline, hyperparameter tuning, ablation experiments across LoRA ranks.	Evaluation framework, failure mode analysis, Hugging Face model card, training visualisations, and final presentation.

Why this impresses the professor: The prescribed course reading explicitly includes the PEFT survey by Han et al. (2024), meaning this project is grounded in assigned material. Publishing the fine-tuned weights on Hugging Face Hub is a concrete real-world contribution that goes beyond most student projects. The ablation study design demonstrates the research rigour the professor's rubric rewards at the 'Excellent' level.

Recommendation & Decision Framework

Use the table below to choose the best project for your team based on your strengths, interests, and the grade outcome you are optimising for.

Project	Best if your team...	Optimises for...
1. Agentic RAG	Has coding confidence and wants enterprise relevance	Technical depth + business impact
2. Multimodal	Enjoys vision/image work and wants a visually stunning demo	Presentation wow-factor + breadth
3. Benchmark Study	Wants to produce original research findings	Research rigour + research paper synergy
4. Text-to-Video	Wants a niche topic with minimal competition in the class	Differentiation + demo appeal
5. Fine-Tuning	Has GPU access and wants a real artefact to publish	Technical credibility + artefact impact

Our Top Pick: Project 1 (Agentic RAG)

Project 1 is the strongest all-round choice because it covers three course weeks simultaneously, produces a working live demo, generates original evaluation results, and mirrors what leading AI companies are deploying in production today. If your team has solid Python skills, start here.

Runner-up pick: Project 3 (Benchmark Study) if your team wants the highest research paper synergy — the benchmark dataset and findings can be directly reused in the 40%-weighted Research Paper assessment, making it the most efficient choice for your overall grade.

Good luck with your final presentation! Map your project explicitly back to learning outcomes A–D in your slides — the rubric rewards this directly.